Is It Deep Enough?
An analysis into Sac Fly Distance

Team 102
Pierre Aucoin, University of Waterloo, pierre.aucoin28@gmail.com

Abstract

Determining how far a ball has to travel for a sacrifice fly to successfully score the runner from third base has been a question for many baseball fans. In this paper, I create a sac fly line that gives fans a perspective on how far a ball has to travel for a successful sac fly. A logistic regression model is created to determine the probability of a successful sac fly based on the fielder's throwing ability, the distance the ball travels and the *pace per ninety feet* of the runner on third. The fielder's throwing ability is quantified by a novel metric I developed called *sniper*, which provides a value that combines a fielder's throwing strength alongside their accuracy in throws to home plate. With this model, it is now possible to compute the distance the ball must travel to have a specific probability of the corresponding sac fly being successful. A Shiny app is used to demonstrate how this can be displayed on the jumbotron during a baseball game to provide fans valuable information regarding the success of a potential sac fly.